LESSON PLAN

EMI/EMC (14)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	27
Total No. of Lesson Plans	46	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Inductive element like EI bar and Synchro as an error detection element in FCS, torque synchro, control synchro, differential and Resolver synchro
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Inductive element like EI bar and Synchro as an error detection element in FCS, torque synchro, control synchro, differential and Resolver synchro.
5. Specific Objectives	14.2.1.1 Trainee will be able to explain the functions of AFCS.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration of Synchro and Resolver operation; Safety: High voltage precautions; Supervision: Instructor monitors connections
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of the AFCS?
(b) Define an inductive element in the context of FCS.
(c) List the types of synchros covered in this lesson.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Error detection using EI bar • Torque synchro construction and operation • Control synchro signal transmission • Differential synchro function	L & Ds	PPT, IB	15 min
(b)	• Resolver synchro principle and application • Comparison of Synchro and Resolver • Integration in FCS error detection loop	L & Ds	PPT, PDF	20 min
(c)	• Troubleshooting inductive element failures • Maintenance requirements for Synchros • Summary of AFCS inductive functions	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how a torque synchro detects position error.
(b) Differentiate between a control synchro and a differential synchro.
(c) Describe the role of the Resolver in an FCS.

12. Home Assignment and Follow Up	Prepare a brief note on how the concepts of Synchro and Resolver discussed in this lesson will be utilized in the signal processing channel covered in LP 02.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Functions of signal processing elements in a FCS, signal processing channel and its elements, limiting synchronizing, gain and adaptive control, feedback in a FCS.	Refer to the master syllabus reference list	LP 28

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min